# Adrenal

## ADRENAL CASE REPORTS I

### *Bilateral Adrenal Hemorrhage with Clinically Preserved Adrenal Function Leading to the Diagnosis of Antiphospholipid Syndrome.*

*Carolina Hurtado, MD, Sarah Sangnim Rhee Kim, MD, Shira Grock, md.*
UCLA, Los Angeles, CA, USA.

**SAT-182**

Background: Spontaneous bilateral adrenal hemorrhage (BAH) is a rare complication of antiphospholipid syndrome (APS), which is the most common identifiable risk factor for BAH. Although adrenal dysfunction is generally irreversible, adrenal function might be preserved or even recover in rare cases[1].

Clinical case: A 48 year-old man with history of hypertension and gout presented with right upper quadrant abdominal pain following trauma to his left leg. He was found to have a left lower extremity deep vein thrombosis and bilateral pulmonary emboli (PE) and was started on anticoagulation therapy. He continued to have abdominal pain and a CT abdomen revealed BAH. Three am cortisol level was 21 mcg/dL (8–25 mcg/dL), ACTH 37 pg/mL (6–59 pg/mL), aldosterone <3 ng/dL (4–31 ng/dL), renin 2.6 ng/mL/hr (0.2–1.6 ng/mL/hr), sodium 130 mmol/L (135–146 mmol/L) and potassium 4.3 mmol/L (3.6–5.3 mmol/L). Patient was hemodynamically stable and did not report symptoms of adrenal insufficiency. Hypercoagulable work-up was consistent with APS and Lupus. Despite normal cortisol levels, he was started on hydrocortisone in the setting of anticoagulation and recent hemorrhage. Given low aldosterone with slightly high renin he was also started on fludrocortisone. Six weeks after discharge, his morning cortisol was 6 mcg/dL and ACTH was elevated at 76 pg/mL which was concerning for adrenal insufficiency. However, 250 mcg IM ACTH stimulation test showed peak cortisol of 17 mcg/dL which is considered adequate. Aldosterone and renin levels normalized so fludrocortisone was discontinued.

Patient subsequently self-discontinued all medications for 1 month with no symptoms of adrenal insufficiency, and later restarted hydrocortisone on his own. Repeat ACTH stimulation test showed baseline ACTH 57 pg/mL with peak cortisol of 17 mcg/dL. Patient was tapered off hydrocortisone and displayed no subsequent symptoms of adrenal insufficiency.

Conclusion: This case highlights the need to consider APS in patients with spontaneous BAH. Additionally, patients with BAH may have relatively preserved adrenal function. There is limited data to guide when steroid replacement is necessary for patients without clear adrenal insufficiency. It may be reasonable to monitor these patients off hydrocortisone replacement with close monitoring.

1. Ramon I, Mathian A, Bachelot A, et al. Primary adrenal insufficiency due to bilateral adrenal hemorrhage-adrenal infarction in the antiphospholipid syndrome: long-term outcome of 16 patients. *J Clin Endocrinol Metab*. 2013;98(8):3179–3189.

# Diabetes Mellitus and Glucose Metabolism

## DIABETES DIAGNOSIS, TREATMENT AND COMPLICATIONS

### *Survival of Patients with Gastroenteropancreatic Neuroendocrine Tumors and Diabetes Mellitus in a Seer-Medicare Cohort*

*Sahityasri Thapi, BA, Kiwoon Baeg, BA, Michelle K. Kim, MD, Emily Jane Gallagher, MB,BCh,BAO, PhD.*
Icahn School of Medicine at Mount Sinai, New York, NY, USA.

**SUN-626**

*Background:* The incidence and prevalence of gastroenteropancreatic neuroendocrine tumors (GEP-NET) is increasing globally and has been associated with diabetes mellitus (DM). In this study we aimed to compare tumor characteristics, disease-specific survival (DSS) and overall survival (OS) of GEP-NET patients (pts) with and without DM. *Methods:* Using the Surveillance, Epidemiology, and End Results registry (SEER) linked to Medicare claims, we identified pts diagnosed with GEP-NET between January 1995 and December 2010, aged ≥65 years at the time of GEP-NET diagnosis. We included patients who were in exclusive Medicare coverage without healthcare management organizations and had Medicare Parts A and B coverage for ≥1year after GEP-NET diagnosis or until death. Within the pts with GEP-NET diagnosis, we identified those without a diagnosis of DM prior to the GEP-NET diagnosis. We compared baseline sociodemographics, co-morbidities, and GEP-NET location, stage, grade and treatment between pts with and without DM using $\chi^2$ analysis. Kaplan Meier (KM) curves were used to compare OS and DSS up to 10 years between the DM and non-DM groups. We used Cox proportional hazards analysis to compare the DSS between the groups, adjusting for confounding variables. *Results:* We identified a cohort of 1,969 well-characterized GEP-NET patients with accurate tumor stage, grade, comorbidities, and treatment data. 478 (25.7%) had DM and 1,383 (74.3%) did not have DM. There were no statistically significant differences in gender or age at the time of GEP-NET diagnosis in the DM (mean age 74.7±SD 6.6 yrs) and non-DM (74.9±7.4 yrs) groups. Significant differences in race were found in the DM (80.6% white, 13.6% black, 1.3% hispanic) and non-DM (86.8% white, 8.2% black, 1.8% Hispanic) groups (p=0.002). Patients with DM had more gastric (14.7%), duodenal (10.9%) and pancreatic (21.0%), and less jejunal/ ileal (12.8%) NETs compared with the non-DM group (9.7%, 6.4%, 16.9%, 18.2%, respectively, p<0.0001). Patients with DM had earlier stage disease than those without DM (p=0.0012), but no difference in tumor grade or treatment was found. KM curves revealed no differences in OS and DSS in the GEP-NET patients with and without DM across all stages. Multivariate adjusted Cox proportional-hazards model found no significant difference in DSS between those with and without DM (HR=0.97, 95%CI: 0.76–1.24). Compared with pts with pancreatic NETs, pts with colon (HR=1.39, 95%CI: 1.04–1.86) had worse survival, while those with jejunal/ileal (HR = 0.59, 95%CI: 0.42–0.83) NETs had a better survival. *Discussion:* This is the first study to investigate the effect of DM on